

Laravel Project Installation Guide

📖 System Requirements

- PHP ≥ 8.1
- Composer (latest version)
- MySQL or compatible database (e.g., MariaDB)
- Node.js & npm (latest stable versions)
- Laravel ≥ 10.x

📖 Installation Steps

1📖 Clone the Repository

Open your terminal and run:

```
git clone https://github.com/yourusername/your-laravel-project.git
cd your-laravel-project
```

2📖 Install Dependencies

Run the following commands inside the project directory:

```
composer install
npm install
npm run build
```

3📖 Configure Environment File

Copy the example environment file and generate an application key:

```
cp .env.example .env
php artisan key:generate
```

Then update your .env file with database credentials:

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=your_database_name
DB_USERNAME=your_username
DB_PASSWORD=your_password
```

4📖 Run Database Migrations

Set up your database tables and seed initial data:

```
php artisan migrate  
php artisan migrate --seed
```

5? Start Background Services (Optional)

Run the queue worker and scheduler:

```
php artisan queue:work  
php artisan schedule:run
```

6? Launch the Application

Start the Laravel development server:

```
php artisan serve
```

7? Access the Application

Open your browser and go to: `http://127.0.0.1:8000`

? Additional Tips

To clear cache and optimize:

```
php artisan optimize:clear  
php artisan optimize
```

To install new dependencies later:

```
composer update  
npm run dev
```

? System Features

1? User Registration (Voters Only)

Fields

- First Name
- Last Name
- Email
- Department
- Student ID
- Profile Image
- Gender (Dropdown)
- Password (Minimum 8 characters)
- Confirm Password (Re-enter Password)

Validations

- Passwords must match and be securely encrypted before storing.
- Prevent duplicate Email or Student ID entries (voters only).
- "Register" button remains disabled until all required fields are filled.

On Success

- Redirect to the registration page with a confirmation message.

2? Login System

Login Types

- Voter Login
- Election Officer Login (for admin-approved users only)

Features

- Forgot Password: Password reset link sent via email.
- Validation: Display error messages for incorrect credentials or unauthorized access.

3? Voter Dashboard

Features

- Profile Management: View and update Department and Profile Image.

- Self-Enrollment: Submit request to become a Candidate or Election Officer.
- Account Management: Delete voter account (without affecting historical election data).
- Access Sections via Icons:
 - - Ongoing Elections – Participate in active elections.
 - - Results – View and download past election reports (PDF).

Voting Process

- OTP (One-Time Password) sent via email for vote confirmation (valid for 15 minutes).
- Prevent duplicate votes per election.
- All votes are encrypted for confidentiality and security.

4 Election Officer Dashboard

Create Elections

- Define Election Name, Date, Positions, and Voting Period.
- Assign Eligible Voters and Candidates (one position per candidate).
- Set Department eligibility for each election.
- Include a “None of the Above (NOTA)” option for every position.

Monitor Elections

- View real-time voting statistics (aggregate data only).
- Election Officers cannot view individual vote details.
- System automatically publishes election results as a PDF report after voting ends.

5 System Admin Dashboard

Authentication

- Access via predefined superuser credentials.

Admin Functions

- User Management: Approve/remove Election Officer requests and delete voter accounts.
- System Management: View/download Audit Logs and perform Database Backup & Recovery.
- Reporting Tools: Generate Custom Reports in PDF or Excel format.